A GEOMETRIC APPROACH TO THE DENSITY OF RANK-METRIC CODES

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Abstract. We study the asymptotic density of linear sections of geometrically irreducible varieties over finite fields that have no $\mathbb{F}_q$-points. We then apply these results to rank-metric codes via determinantal varieties. Our approach recovers known results in the sparse and abundant cases with concise proofs, and also settles the threshold case. To address the latter, we introduce the notion of quasireflexive varieties and show that determinantal varieties satisfy this property. This allows us to invoke the Chebotarev density theorem for varieties over finite fields to obtain the desired estimate.

1. Introduction

Let $\mathbb{F}_q$ be the finite field with q elements, where q is a prime power, and regard $\mathbb{F}_q^{mn}$ as the space of $m \times n$ matrices over $\mathbb{F}_q$ equipped with the rank metric

$$d(A, B) := \mathrm{rank}(A - B), \qquad A, B \in \mathbb{F}_q^{mn}.$$

Throughout the paper, we assume without loss of generality that $m \geq n$. A *linear rank-metric code* is a linear subspace $V \subseteq \mathbb{F}_q^{mn}$. One of the most important invariants of such a code is its *minimum (rank) distance*, defined by

$$\begin{aligned} d_R(V) &:= \min \{d(A, B) \mid A, B \in V,\ A \neq B\} \\ &= \min \{\mathrm{rank}(M) \mid M \in V \setminus \{0\}\}. \end{aligned}$$

The minimum distance determines the error-correcting capabilities of the code.

Rank-metric codes have become a central object of study in modern coding theory. They arise naturally in contexts where errors are measured by rank, with important applications to random network coding and secure network coding [SKK08, SK11]. They are also closely related to several areas of pure mathematics, including q-polymatroids and finite geometry [GJLR20, CMPZ17]. For a general introduction to the subject, we refer the reader to the surveys [GKR23, BHL+22].

The main problem that we study is motivated by analogous results for linear codes in the Hamming metric. There has been extensive study of the distribution of the minimum distance among all k-dimensional linear codes $C \subseteq \mathbb{F}_q^n$. For a recent notable example, see [HHLT22]. Much of the work in this area has focused on the asymptotic problem where q is fixed, while k and n grow together in such a way that k/n converges. Loidreau studies the analogue of this problem for rank-metric codes in [Loi14].

We are interested in a different kind of limit where q tends to infinity while the other parameters remain fixed. Recall that if $C \subseteq \mathbb{F}_q^n$ is a k-dimensional linear code with minimum Hamming distance d, then the Singleton bound gives $d \leq n-k+1$. Codes for which equality holds are called *MDS* (maximum distance separable). When $q \to \infty$ with k and n fixed, it is not difficult to show that most k-dimensional linear codes $C \subseteq \mathbb{F}_q^n$ are MDS. For a quantitative result in this direction, see [Kai14].

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We phrase our main results in terms of the following density. Fix integers k and δ, and consider the density of k-dimensional codes with minimum rank distance at least δ:

$$D_q(m\times n,k,\delta) := \frac{\left|\{\text{codes } V\subseteq \mathbb{F}_q^{mn} \mid \dim V = k,\ d_R(V)\geq \delta\}\right|}{\left|\{\text{codes } V\subseteq \mathbb{F}_q^{mn} \mid \dim V = k\}\right|}.$$

The rank-metric analogue of the Singleton bound asserts that for a k-dimensional linear rank-metric code $V\subseteq \mathbb{F}_q^{mn}$, we have $k\leq m(n-d_R(V)+1)$ [GKR23, Theorem 3.1]. Codes that reach equality in this bound are called *MRD* (maximum rank distance). They are the rank-metric analogues of MDS codes.

There has been significant recent interest in density questions for MRD codes. For an overview of this topic, see [GKR23, Section 6]. Gruica and Ravagnani have determined the asymptotic density of MRD codes for all sets of parameters.

Theorem 1.1. [GR22, Theorem 5.9] *Assume that* $1\leq\delta\leq n$. *Then*

$$\lim_{q\to\infty} D_q(m\times n,\, m(n-\delta+1),\, \delta) = \begin{cases} 1 & \text{if } \delta = 1,\\ \sum_{i=0}^m \frac{(-1)^i}{i!} & \text{if } n=\delta=2,\\ 0 & \text{otherwise.}\end{cases}$$

The $n=\delta=2$ case is due to Antrobus and Gluesing-Luerssen [AGL19, Corollary VII.5].

Gruica and Ravagnani apply their techniques to obtain upper and lower bounds on the density of rank-metric codes with more general minimum distance and dimension. This allows them to determine the behavior of $D_q(m\times n,k,\delta)$ as $q\to\infty$ in most cases.

Theorem 1.2. [GR22, Theorem 5.12] *We have*

$$\lim_{q\to\infty} D_q(m\times n,k,\delta) = \begin{cases} 0 & \text{if} \quad (\delta-1)(m+n-\delta+1) > mn-k+1,\\ 1 & \text{if} \quad (\delta-1)(m+n-\delta+1) < mn-k+1.\end{cases}$$

In view of this result, we refer to the first situation as the *sparse case* and the second as the *abundant case*. The remaining situation is the *threshold case*, namely when

$$(\delta-1)(m+n-\delta+1) = mn-k+1.$$

Gruica and Ravagnani explain that in the threshold case, [GR22, Proposition 4.4] implies the upper bound

$$\limsup_{q\to\infty} D_q(m\times n,k,\delta)\leq \frac{1}{2}.$$

They further remark that their techniques do not determine the limit in this case and that, in general, rank-metric codes in this setting are neither sparse nor dense as q grows [GR22, Remark 5.13].

1.1. **Main results.** In this paper, we adopt an algebro-geometric approach to these density questions. Our main theorem (Theorem 1.5) resolves the threshold case as a corollary, and also yields concise proofs for the sparse and abundant cases.

Corollary 1.3. *Fix integers* $m\geq n$, $2\leq\delta\leq n$, *and* $1\leq k\leq mn$, *and set*

$$\Delta := \prod_{i=0}^{n-\delta} \frac{\binom{m+i}{\delta-1}}{\binom{\delta-1+i}{\delta-1}}.$$

Then the asymptotic density of k-dimensional codes in $\mathbb{F}_q^{mn}$ with minimum rank distance at least δ is given by

$$\lim_{q\to\infty} D_q(m\times n,k,\delta) = \begin{cases} 0 & \textit{if} \quad (\delta-1)(m+n-\delta+1) > mn-k+1, \\ 1 & \textit{if} \quad (\delta-1)(m+n-\delta+1) < mn-k+1, \\ \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!} & \textit{if} \quad (\delta-1)(m+n-\delta+1) = mn-k+1. \end{cases}$$

For completeness, when $1 \le k \le mn$, we have $D_q(m\times n,k,1) = 1$, while $D_q(m\times n,k,\delta) = 0$ for every $\delta > n$.

Our main theorem concerns the density of linear sections of determinantal varieties that contain no $\mathbb{F}_q$-rational points. We now introduce the background necessary to state this result precisely. Let δ be an integer satisfying $2 \le \delta \le n$, and set $W := \mathrm{Mat}_{m\times n}(\mathbb{F}_q)$. Inside $\mathbb{P}(W) \cong \mathbb{P}^{mn-1}_{\mathbb{F}_q}$, let Y_δ be the projective determinantal variety cut out by the $\delta\times\delta$ minors. More concretely, we can describe the points of Y_δ as follows:

$$Y_\delta = \left\{ [M] \in \mathbb{P}^{mn-1} \mid \mathrm{rank}(M) < \delta \right\}.$$

By definition, a code $V \subseteq \mathbb{F}_q^{mn}$ has $d_R(V) \ge \delta$ if and only if the linear section $\mathbb{P}(V)\cap Y_\delta$ contains no $\mathbb{F}_q$-point. Moreover, [Har92, Proposition 12.2] gives

$$\begin{aligned} \dim Y_\delta &= (mn-1) - (m-\delta+1)(n-\delta+1) \\ &= (\delta-1)(m+n-\delta+1) - 1. \end{aligned}$$

The dimension of a variety can be characterized by its general linear sections [Har92, Definitions 11.2–11.3]. Accordingly, for a general linear subspace $V \subseteq \overline{\mathbb{F}_q}^{mn}$ of dimension k,

$$\mathbb{P}(V)\cap Y_\delta \quad \text{is} \quad \begin{cases} \text{positive-dimensional} & \text{if} \quad (\delta-1)(m+n-\delta+1) > mn-k+1, \\ \text{empty} & \text{if} \quad (\delta-1)(m+n-\delta+1) < mn-k+1, \\ \text{zero-dimensional} & \text{if} \quad (\delta-1)(m+n-\delta+1) = mn-k+1, \end{cases}$$

which correspond precisely to the sparse, abundant, and threshold cases, respectively.

Determinantal varieties are examples of *geometrically irreducible varieties*, meaning that they remain irreducible in the Zariski topology after extending the base field to the algebraic closure $\overline{\mathbb{F}_q}$; see, for example, [Har92, Proposition 12.2]. They also satisfy *quasireflexivity*, a notion introduced for curves by Entin [Ent21, Section 2]. More precisely, a possibly reducible curve $C \subseteq \mathbb{P}^n$ of degree d is quasireflexive if, for every irreducible component C_i of C, there exists a hyperplane tangent to C_i whose set-theoretic intersection with C consists of $d-1$ geometric points. Thus, the tangency is as simple as possible, while all other intersections are transverse. We extend this notion to higher dimensions as follows.

Definition 1.4. A geometrically irreducible projective variety $Y \subseteq \mathbb{P}^N$ of degree Δ is *quasireflexive* if there exists a linear subspace $L \subseteq \mathbb{P}^N$ of dimension $N - \dim Y$ whose intersection with Y consists of $\Delta - 1$ points at which Y is smooth. More generally, a pure-dimensional projective variety is quasireflexive if, for each of its geometrically irreducible components, there exists such an L whose tangency point lies on that component.

Quasireflexivity enables us to apply the *Chebotarev density theorem*; see Section 3 for further details. The main result in this paper is as follows.

Theorem 1.5. *Let $Y \subseteq \mathbb{P}^N$ be a geometrically irreducible variety over $\mathbb{F}_q$ of degree Δ, and let $\mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)$ be the Grassmannian of ℓ-dimensional linear subspaces in $\mathbb{P}^N$. Consider the density function*

$$D_q(Y,\ell) = \frac{\left|\left\{L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \;\middle|\; (L \cap Y)(\mathbb{F}_q) = \emptyset\right\}\right|}{|\mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q)|}.$$

(1) *There exists a constant c_1 depending only on N, Δ, ℓ such that*
$$0 \leq D_q(Y,\ell) \leq c_1 q^{N-\dim Y-\ell}.$$
In particular, the condition $\dim Y + \ell > N$ implies
$$\lim_{q\to\infty} D_q(Y,\ell) = 0.$$

(2) *There exists a constant c_2 depending only on N, Δ, ℓ such that*
$$1 - c_2 q^{\dim Y+\ell-N} \leq D_q(Y,\ell) \leq 1.$$
In particular, the condition $\dim Y + \ell < N$ implies
$$\lim_{q\to\infty} D_q(Y,\ell) = 1.$$

(3) *Suppose $\dim Y + \ell = N$ and Y is quasireflexive. Then*
$$\lim_{q\to\infty} D_q(Y,\ell) = \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}.$$

We prove that determinantal varieties are quasireflexive in Theorem 4.12. To compute the asymptotic density of rank-metric codes, we apply Theorem 1.5 with

$$Y = Y_\delta \subseteq \mathbb{P}^{mn-1}, \qquad N = mn-1, \qquad \ell = k-1.$$

The degree formula for determinantal varieties (see [Ful98, Example 14.4.11]) gives

$$\Delta = \deg(Y_\delta) = \prod_{i=0}^{n-\delta} \frac{\binom{m+i}{\delta-1}}{\binom{\delta-1+i}{\delta-1}}.$$

In this setting, we have $D_q(m\times n, k, \delta) = D_q(Y_\delta,\, k-1)$. Thus, Theorem 1.5 directly yields Corollary 1.3.

1.2. **Related work.** We now discuss how our method relates to previous approaches to density questions for rank-metric codes. Byrne and Ravagnani developed a combinatorial approach based on families of codes that are balanced with respect to a given partition of the ambient space [BR20]. For MRD codes with $n = \delta = 2$, Antrobus and Gluesing-Luerssen utilized enumerative results on square matrices over $\mathbb{F}_q$ with no eigenvalue in $\mathbb{F}_q$ [AGL19]. Gluesing-Luerssen deduced an exact formula for $D_q(3\times 3, 3, 3)$ via the connection between square full-rank MRD codes and the theory of semifields [GL20, Theorem 2.4]. This connection was further developed by Gruica, Ravagnani, Sheekey, and Zullo in [GRSZ23, Section 3].

Gruica and Ravagnani [GR22] study density questions by counting common complements of collections of linear subspaces. As explained in [GKR23, Remark 6.6 (i)], for each $m \times n$ matrix $M \in \mathbb{F}_q^{mn}$ with $\mathrm{rank}(M) < \delta$, its column space is contained in some $(\delta-1)$-dimensional subspace $U \subseteq \mathbb{F}_q^m$. For a fixed such U, the matrices whose column spaces are contained in U form a subspace $\mathbb{F}_q^{mn}(U) \subseteq \mathbb{F}_q^{mn}$. The MRD codes of dimension k and minimum distance δ correspond to common complements of the spaces $\mathbb{F}_q^{mn}(U)$, where U ranges over all $(\delta-1)$-dimensional subspaces of $\mathbb{F}_q^m$. The problem of estimating the number of such common complements can then be turned into one of counting isolated vertices in a certain bipartite graph. This approach fits into the study of the Critical Problem for combinatorial geometries. For a more extensive discussion of this connection, see [GRSZ23, Section 5].

In some ways, our geometric approach using linear sections of determinantal varieties is similar to the common-complement approach described above. The key distinction is that, rather than viewing the determinantal variety of matrices of rank at most $\delta-1$ as a union of finitely many subspaces, we treat it as a single irreducible projective variety, exploiting more of the geometry inherent in the problem.

The previous work most closely related to our approach concerns a different kind of limit. In [NHTRR18], Neri, Horlemann-Trautmann, Randrianarisoa, and Rosenthal study the proportion of $\mathbb{F}_{q^m}$-linear MRD codes as the field extension degree m tends to infinity. A linear MRD code $C \subseteq \mathbb{F}_{q^m}^n$ of dimension k has a $k \times n$ generator matrix G with entries in $\mathbb{F}_{q^m}$ of the form $G = (I_k \mid X)$, which is called *systematic form* or *standard form*. Building on a general criterion for generator matrices of MRD codes [NHTRR18, Proposition 13], the authors show that the $k \times (n-k)$ matrices X yielding an MRD code form a Zariski open subset of the space of all such matrices [NHTRR18, Theorem 17]. Whereas they work in an affine space of matrices, we take advantage of the nicer geometric properties of the corresponding Grassmannian in projective space.

We end the introduction with several directions not pursued in this paper:

Rate of convergence. Corollary 1.3 computes the limit of $D_q(m \times n, k, \delta)$ as $q \to \infty$, but it does not address the rate of convergence. As an illustrative example, in the case where $n = m = k = \delta = 3$, our arguments yield an upper bound of the form $O(q^{-1})$ for the density, whereas Gluesing-Luerssen, using connections to the theory of semifields, obtained an exact formula that is $O(q^{-3})$ [GL20, Theorem 2.4]. Although Theorem 1.5 provides some information about the rate of convergence, we have not attempted to optimize it here.

Field of linearity. After fixing a basis of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$, each element of $\mathbb{F}_{q^m}$ can be identified with a vector in $\mathbb{F}_q^m$, and hence each element of $\mathbb{F}_{q^m}^n$ with an $m \times n$ matrix over $\mathbb{F}_q$. Thus, if ℓ divides m, an $\mathbb{F}_{q^\ell}$-linear subspace of $\mathbb{F}_{q^m}^n$ corresponds to an $\mathbb{F}_q$-linear subspace of the space of $m \times n$ matrices. Note that every $\mathbb{F}_{q^\ell}$-linear code is $\mathbb{F}_q$-linear, whereas the converse does not hold in general. This issue is discussed explicitly in [GHRW24, Remark 5.2].

There has been considerable interest in density questions for such $\mathbb{F}_{q^\ell}$-linear rank-metric codes. In [GHRW24], Gruica, Horlemann, Ravagnani, and Willenborg investigate these questions in a broader setting. In particular, they consider an analogue of $D_q(m \times n, k, \delta)$ with an additional parameter specifying that only $\mathbb{F}_{q^\ell}$-linear subspaces are counted. They also study analogous density questions for codes endowed with other metrics, including the sum-rank metric.

Other limits of interest. In [NHTRR18], Neri, Horlemann-Trautmann, Randrianarisoa, and Rosenthal focus on the $\mathbb{F}_{q^m}$-linear rank-metric codes described above and show that $\mathbb{F}_{q^m}$-linear MRD codes are dense as $m \to \infty$. This result was generalized by Gruica, Horlemann, Ravagnani, and Willenborg, who proved that $\mathbb{F}_{q^\ell}$-linear MRD codes in $\mathbb{F}_{q^m}^n$ are dense as $\ell \to \infty$, where ℓ divides m [GHRW24, Theorem 5.8].

Our geometric arguments, by contrast, require the full set of k-dimensional $\mathbb{F}_q$-linear subspaces of the space of $m \times n$ matrices over $\mathbb{F}_q$, rather than only those that are linear over some extension field $\mathbb{F}_{q^\ell}$. We also require the dimension of the matrix space to remain fixed as $q \to \infty$. Consequently, our results do not address other limits of interest, such as the limit of $D_q(m \times n, k, \delta)$ as $m \to \infty$.

Organization of the paper. The first two cases of Theorem 1.5 are proved in Section 2, while the remaining threshold case is treated in Section 3. In Section 4, we verify that determinantal varieties are quasireflexive. In Appendix A, we discuss the closely related notion of reflexivity, and prove that determinantal varieties satisfy this property in characteristic different from 2.

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2. Asymptotic density in the sparse and abundant cases

We prove the sparse and abundant cases of Theorem 1.5 in this section.

2.1. **The sparse case.** For any projective variety $Y \subseteq \mathbb{P}^N$, choose $L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q)$ uniformly at random and define

$$\begin{aligned}\Theta\colon \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) &\longrightarrow \mathbb{Z}_{\geq 0}\\ L &\longmapsto |(L \cap Y)(\mathbb{F}_q)|.\end{aligned}$$

We view Θ as a random variable. As shown for hyperplanes in [PS22, Lemma 4.1] and for higher-codimensional linear sections in [KS22, Lemma 2.1], the mean μ and variance σ^2 of Θ satisfy

$$\mu = |Y(\mathbb{F}_q)| \left(q^{\ell-N} + O\left(q^{\ell-N-1}\right)\right), \qquad \sigma^2 = O\left(q^{\ell-N}|Y(\mathbb{F}_q)|\right).$$

If Y is geometrically irreducible, the Lang–Weil bound [LW54, Theorem 1] gives

$$|Y(\mathbb{F}_q)| = q^{\dim Y} + O\left(q^{\dim Y - \frac{1}{2}}\right).$$

Substituting this into the preceding estimates, we obtain

$$\mu = q^{\dim Y+\ell-N} + O\left(q^{\dim Y+\ell-N-\frac{1}{2}}\right), \qquad \sigma^2 = O\left(q^{\dim Y+\ell-N}\right). \tag{2.1}$$

Proof of Theorem 1.5 (1). The inequality $D_q(Y,\ell) \geq 0$ holds by definition, so it remains to show that there exists a constant c_1, depending only on N, Δ, ℓ, such that

$$D_q(Y,\ell) \leq c_1 q^{N-\dim Y-\ell}.$$

Consider the set of pointless linear sections

$$\mathcal{U}_q := \left\{L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \;\middle|\; (L \cap Y)(\mathbb{F}_q) = \emptyset\right\}.$$

For any $L \in \mathcal{U}_q$, we have $|(L \cap Y)(\mathbb{F}_q)| = 0$. Hence,

$$||(L \cap Y)(\mathbb{F}_q)| - \mu| = \mu \geq \mu = \left(\frac{\mu}{\sigma}\right)\sigma.$$

It follows that

$$\mathrm{Prob}\left(L \in \mathcal{U}_q\right) \leq \mathrm{Prob}\left(||(L \cap Y)(\mathbb{F}_q)| - \mu| \geq \left(\frac{\mu}{\sigma}\right)\sigma\right)$$

$$
\begin{aligned}
&\text{(Chebyshev's inequality)} && \leq \left(\frac{\sigma}{\mu}\right)^2 \\
&\text{(Equation (2.1))} && = O\left(q^{N-\dim Y-\ell}\right).
\end{aligned}
$$

This establishes the desired upper bound. In particular, the condition $\dim Y + \ell > N$ implies $\lim_{q\to\infty} q^{N-\dim Y-\ell} = 0$, which further implies $\lim_{q\to\infty} D_q(Y,\ell) = 0$. $\square$

Remark 2.1. Our proof for the limit $\lim_{q\to\infty} D_q(Y,\ell) = 0$ shows more precisely that

$$D_q(Y,\ell) = O(q^{N-\dim Y-\ell}).$$

In the sparse case, $\dim Y + \ell > N$, so the exponent is negative. This gives a quantitative upper bound on the rate at which $D_q(Y,\ell)$ tends to zero, although the bound need not be sharp. For instance, the determinantal variety $Y_3 \subseteq \mathbb{P}^8$ of 3×3 matrices of rank < 3 (that is, singular matrices) has dimension 7. With $\ell = 2$, the above estimate gives $D_q(Y,2) = O(q^{-1})$, whereas Gluesing-Luerssen obtained the sharper bound $O(q^{-3})$ in this case using semifield theory [GL20, Theorem 2.4].

2.2. **The abundant case.** Consider a geometrically irreducible variety $Y \subseteq \mathbb{P}^N$ defined over $\mathbb{F}_q$, together with the incidence correspondence

$$\mathcal{X} := \left\{(y,L) \in Y \times \mathrm{Gr}(\mathbb{P}^\ell,\mathbb{P}^N) \;\middle|\; y \in L\right\}.$$

There are two natural projection maps

$$
\begin{tikzcd}
 & \mathcal{X} \arrow[dl, "\pi_Y"'] \arrow[dr, "\pi_{\mathrm{Gr}}"] & \\
Y & & \mathrm{Gr}(\mathbb{P}^\ell,\mathbb{P}^N),
\end{tikzcd}
$$

where the image of the second projection is

$$\mathcal{N} := \pi_{\mathrm{Gr}}(\mathcal{X}) = \left\{L \in \mathrm{Gr}(\mathbb{P}^\ell,\mathbb{P}^N) \;\middle|\; L \cap Y \neq \emptyset\right\}.$$

To clarify the (geometric) points of $\mathcal{N}$, observe that

$$\mathcal{N}(\overline{\mathbb{F}_q}) = \left\{L \in \mathrm{Gr}(\mathbb{P}^\ell,\mathbb{P}^N)(\overline{\mathbb{F}_q}) \;\middle|\; L \cap Y_{\overline{\mathbb{F}_q}} \neq \emptyset\right\}. \tag{2.2}$$

Lemma 2.2. *The incidence correspondence $\mathcal{X}$ is geometrically irreducible with*

$$\dim \mathcal{X} = \dim \mathrm{Gr}(\mathbb{P}^\ell,\mathbb{P}^N) - (N - \dim Y - \ell).$$

Proof. If $\ell = 0$, then $\mathrm{Gr}(\mathbb{P}^0,\mathbb{P}^N) \cong \mathbb{P}^N$, and $\mathcal{X}$ is the graph of the inclusion $Y \hookrightarrow \mathbb{P}^N$. Since $\mathcal{X} \cong Y$, the assertion follows. For the remainder of the proof, we assume that $\ell \geq 1$.

For any $y \in Y$, the fiber $\pi_Y^{-1}(y)$ consists of ℓ-dimensional linear subspaces of $\mathbb{P}^N$ containing y. Projecting from y identifies such subspaces with $(\ell-1)$-dimensional linear subspaces of $\mathbb{P}^{N-1}$, yielding an isomorphism

$$\pi_Y^{-1}(y) \cong \mathrm{Gr}(\mathbb{P}^{\ell-1},\mathbb{P}^{N-1}).$$

Hence, the fibers of π_Y are geometrically irreducible and have the same dimension. Since Y is geometrically irreducible, it follows that $\mathcal{X}$ is geometrically irreducible by [Sha13, Theorem 1.26].

For the dimension, we compute

$$\dim \mathcal{X} = \dim \pi_Y^{-1}(y) + \dim Y = \dim \mathrm{Gr}(\mathbb{P}^{\ell-1},\mathbb{P}^{N-1}) + \dim Y$$

$$
\begin{aligned}
&= \ell(N-\ell) + \dim Y \\
&= \ell(N-\ell) + (N-\ell) + \dim Y - (N-\ell) \\
&= (\ell+1)(N-\ell) + \dim Y - (N-\ell) \\
&= \dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) - (N - \dim Y - \ell).
\end{aligned}
$$

This proves the statement. $\square$

Lemma 2.3. *The projection image $\mathcal{N} = \pi_{\mathrm{Gr}}(\mathcal{X})$ is geometrically irreducible with*

$$\dim \mathcal{N} \leq \dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) - (N - \dim Y - \ell).$$

Proof. The variety $\mathcal{N}$ is geometrically irreducible since $\mathcal{X}$ is geometrically irreducible by Lemma 2.2. The upper bound on the dimension follows from the inequality $\dim \mathcal{N} \leq \dim \mathcal{X}$ together with Lemma 2.2. $\square$

Proof of Theorem 1.5 (2). By definition,

$$1 - D_q(Y, \ell) = \frac{\left| \left\{ L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \;\middle|\; (L \cap Y)(\mathbb{F}_q) \neq \emptyset \right\} \right|}{\left| \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \right|}.$$

In view of (2.2), observe that $\left\{ L \in \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \;\middle|\; (L \cap Y)(\mathbb{F}_q) \neq \emptyset \right\}$ is a subset of $\mathcal{N}(\mathbb{F}_q)$. This inclusion may be strict, since an $\mathbb{F}_q$-rational linear subspace can meet Y geometrically without meeting $Y(\mathbb{F}_q)$. In any event,

$$1 - D_q(Y, \ell) \leq \frac{|\mathcal{N}(\mathbb{F}_q)|}{\left| \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \right|}. \tag{2.3}$$

The variety $\mathcal{N}$ is geometrically irreducible by Lemma 2.3, so we can apply the Lang–Weil bound [LW54, Theorem 1] to obtain

$$\left| |\mathcal{N}(\mathbb{F}_q)| - q^{\dim \mathcal{N}} \right| \leq A q^{\dim \mathcal{N} - 1/2}$$

where A is a constant depending only on N, Δ, ℓ. From this inequality, one can check that for $q \geq \max\{A^2, 1\}$,

$$|\mathcal{N}(\mathbb{F}_q)| \leq q^{\dim \mathcal{N}} + A q^{\dim \mathcal{N} - 1/2} \leq 2 q^{\dim \mathcal{N}}. \tag{2.4}$$

Next, we observe that

$$\left| \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)(\mathbb{F}_q) \right| = \prod_{j=0}^{\ell} \frac{q^{N+1-j} - 1}{q^{\ell+1-j} - 1} \geq q^{(\ell+1)(N-\ell)} = q^{\dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)}. \tag{2.5}$$

Indeed, each factor is at least $q^{N-\ell}$, since $(q^{N+1-j} - 1) - q^{N-\ell}(q^{\ell+1-j} - 1) = q^{N-\ell} - 1 \geq 0$.

Applying Lemma 2.3 in combination with (2.3), (2.4), and (2.5), we obtain the following estimate for all sufficiently large q:

$$1 - D_q(Y, \ell) \leq \frac{2 q^{\dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) - (N - \dim Y - \ell)}}{q^{\dim \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)}} = 2 q^{\dim Y + \ell - N}.$$

Thus, there exists a constant c_2 depending only on N, Δ, ℓ such that for all q

$$1 - D_q(Y, \ell) \leq c_2 q^{\dim Y + \ell - N}.$$

Rearranging the terms gives

$$1 - c_2 q^{\dim Y + \ell - N} \leq D_q(Y, \ell) \leq 1.$$

If $\dim Y + \ell < N$, then $\lim_{q \to \infty} (1 - c_2 q^{\dim Y + \ell - N}) = 1$, and thus $\lim_{q \to \infty} D_q(Y, \ell) = 1$. $\square$

3. Asymptotic density in the threshold case

Consider a geometrically irreducible variety $Y \subseteq \mathbb{P}^N$ of degree Δ over $\mathbb{F}_q$, together with the incidence correspondence

$$
\begin{tikzcd}
& \mathcal{X} := \left\{(y, L) \in Y \times \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N) \;\middle|\; y \in L\right\} \arrow[dl, "\pi_Y"'] \arrow[dr, "\pi_{\mathrm{Gr}}"] & \\
Y & & \mathcal{G} := \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N).
\end{tikzcd}
$$

Assume that $\dim Y + \ell = N$. Then a general ℓ-plane $L \in \mathcal{G}$ intersects Y transversely in Δ points by Bertini's theorem. In particular, the map

$$\pi_{\mathrm{Gr}} \colon \mathcal{X} \longrightarrow \mathcal{G}$$

is generically finite of degree Δ. Consequently, there exists a nonempty Zariski open subset $\mathcal{G}^\circ \subseteq \mathcal{G}$ over which the restriction of π_{Gr} is finite étale of degree Δ. In this setting, Theorem 1.5 (3) concerns the distribution of $\mathbb{F}_q$-points $L \in \mathcal{G}(\mathbb{F}_q)$ for which the fiber $\pi_{\mathrm{Gr}}^{-1}(L) = L \cap Y$ contains no $\mathbb{F}_q$-points.

3.1. **Chebotarev density theorem.** Before proceeding to the proof, we briefly review the background for the key ingredient. The main source for the following material is [Ent21, Sections 3 and 4]. Let $\mathcal{G}^\circ$ be a geometrically irreducible quasi-projective variety, and let

$$f \colon \mathcal{X}^\circ \longrightarrow \mathcal{G}^\circ$$

be a finite étale morphism of degree Δ. For a geometric point $L_0 \in \mathcal{G}^\circ(\overline{\mathbb{F}_q})$, the étale fundamental group $\pi^{\mathrm{et}}(\mathcal{G}^\circ, L_0)$ acts on the fiber $f^{-1}(L_0)$, giving rise to a homomorphism to the symmetric group S_Δ. The *arithmetic monodromy group* $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$ is defined as the image of this homomorphism. This group is well defined up to conjugation since different choices of the base point L_0 yield conjugate subgroups of S_Δ.

On the other hand, the *geometric monodromy group* is defined by

$$\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) := \mathrm{Mon}\left(\mathcal{X}^\circ \times_{\mathbb{F}_q} \overline{\mathbb{F}_q} / \mathcal{G}^\circ \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}\right).$$

The two monodromy groups fit into a short exact sequence

$$1 \longrightarrow \mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) \longrightarrow \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \longrightarrow \mathrm{Gal}(\mathbb{F}_{q^\nu}/\mathbb{F}_q) \longrightarrow 1,$$

where ν is the minimal positive integer such that

$$\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) = \mathrm{Mon}\left(\mathcal{X}^\circ \times_{\mathbb{F}_q} \mathbb{F}_{q^\nu} / \mathcal{G}^\circ \times_{\mathbb{F}_q} \mathbb{F}_{q^\nu}\right).$$

In particular, the arithmetic and geometric monodromy groups may not coincide, even when q is large. In our application, we will prove directly that the geometric monodromy group is S_Δ. Since the arithmetic monodromy group is a subgroup of S_Δ containing the geometric monodromy group, it will follow that both groups are equal to S_Δ.

For each point $L \in \mathcal{G}^\circ(\mathbb{F}_q)$, the fiber $f^{-1}(L) \subseteq \mathcal{X}^\circ$ is a 0-dimensional scheme over $\mathbb{F}_q$ consisting of Δ geometric points. The action of the Frobenius map on this fiber determines a conjugacy class

$$\mathrm{Frob}_q(L) \subseteq \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ).$$

Note that $f^{-1}(L)$ contains no $\mathbb{F}_q$-points if and only if the class $\mathrm{Frob}_q(L)$ consists of derangements. This idea reduces our problem to estimating the density of points $L \in \mathcal{G}^\circ(\mathbb{F}_q)$ for which $\mathrm{Frob}_q(L)$ consists of derangements.

The key tool for our argument is the following version of the Chebotarev density theorem proved by Entin [Ent21]. We state only the form relevant to our setting, namely the case in which the arithmetic and geometric monodromy groups coincide.

Theorem 3.1 ([Ent21, Theorem 3])**.** *Let $f\colon \mathcal{X}^\circ \to \mathcal{G}^\circ$ be a finite étale morphism between quasiprojective varieties over $\mathbb{F}_q$ with $\mathcal{G}^\circ$ geometrically irreducible. Suppose that*

$$\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) = \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ).$$

Then for each conjugacy class $\mathcal{C} \subseteq \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)$, we have

$$|\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \mid \mathrm{Frob}_q(L) = \mathcal{C}\}| = \frac{|\mathcal{C}|}{|\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ)|} q^{\dim \mathcal{G}^\circ}\left(1 + O\left(q^{-1/2}\right)\right).$$

Remark 3.2. The implied constant in $O\left(q^{-1/2}\right)$ depends on the *complexity* of the morphism f. Roughly speaking, this complexity measures the ambient projective dimension and the degrees of the equations and nonvanishing conditions defining the graph of f; see [Ent21, Section 4]. In our setting, the determinantal variety and the associated incidence constructions are defined uniformly over the base field. In particular, the complexity of f is bounded independently of q.

3.2. **Proof for the threshold case.** We now retain the notation introduced in the beginning of this section. In our application, $\mathcal{G}^\circ \subseteq \mathcal{G}$ is a Zariski open subset over which π_{Gr} is étale, and we set $\mathcal{X}^\circ := \pi_{\mathrm{Gr}}^{-1}(\mathcal{G}^\circ)$. The restriction

$$f := \pi_{\mathrm{Gr}}|_{\mathcal{X}^\circ} \colon \mathcal{X}^\circ \longrightarrow \mathcal{G}^\circ$$

is therefore a finite étale morphism of degree Δ.

Lemma 3.3. *If the variety Y is quasireflexive, then* $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \cong S_\Delta$.

Proof. The assertion is immediate when $\Delta = 1$, so assume that $\Delta \geq 2$. Fix a geometric point $L_0 \in \mathcal{G}^\circ(\mathbb{F}_q)$ and identify $\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ)$ as a subgroup of S_Δ through its action on the geometric fiber $f^{-1}(L_0)$. Let

$$D \subseteq \mathcal{X} \times_{\mathcal{G}} \mathcal{X} \qquad \text{and} \qquad D' \subseteq Y \times Y$$

denote the diagonals. The fiber of the morphism

$$\mathcal{X} \times_{\mathcal{G}} \mathcal{X} \setminus D \longrightarrow Y \times Y \setminus D'$$

over $(y_1, y_2) \in Y \times Y \setminus D'$ consists of the points $((y_1, L), (y_2, L))$, where $L \in \mathcal{G} = \mathrm{Gr}(\mathbb{P}^\ell, \mathbb{P}^N)$ contains the line spanned by y_1 and y_2. This realizes $\mathcal{X} \times_{\mathcal{G}} \mathcal{X} \setminus D$ as a Grassmannian bundle over the irreducible variety $Y \times Y \setminus D'$, so it is itself irreducible. It follows that $\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ)$ acts doubly transitively on $f^{-1}(L_0)$ by [BH86, Proposition 2 (ii)].

It remains to produce a simple transposition. Since Y is quasireflexive, there exists $L \in \mathcal{G}$ meeting the smooth locus of Y in $\Delta - 1$ points $y_0, y_1, \dots, y_{\Delta-2}$, where y_0 has multiplicity two and all other intersections are transverse. The map π_{Gr} is étale at $(y_i, L) \in \mathcal{X}$ for all $i = 1, \dots, \Delta-2$. Moreover, $\mathcal{X}$ is locally irreducible at (y_0, L), since it is a Grassmannian bundle over Y and Y is smooth at y_0. Therefore, $\mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ)$ contains a simple transposition by [BH86, Proposition 3], and hence is isomorphic to S_Δ. Finally, the chain

$$S_\Delta = \mathrm{Mon}^g(\mathcal{X}^\circ/\mathcal{G}^\circ) \subseteq \mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \subseteq S_\Delta$$

shows that equality must hold throughout, which proves that $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \cong S_\Delta$. ∎

Lemma 3.4. *If the variety Y is quasireflexive, then, as $q \to \infty$,*

$$\left|\left\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \;\middle|\; \mathrm{Frob}_q \text{ deranges } \pi_{\mathrm{Gr}}^{-1}(L)\right\}\right| = \left(\sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}\right) q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right).$$

Proof. By Lemma 3.3, we have $\mathrm{Mon}(\mathcal{X}^\circ/\mathcal{G}^\circ) \cong S_\Delta$. Applying Theorem 3.1, we obtain

$$\begin{aligned}\left|\left\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \;\middle|\; \mathrm{Frob}_q \text{ deranges } \pi_{\mathrm{Gr}}^{-1}(L)\right\}\right| &= \sum_{\substack{\mathcal{C} \text{ consists of} \\ \text{derangements}}} \frac{|\mathcal{C}|}{|S_\Delta|} q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right) \\ &= \frac{|\{\text{derangements in } S_\Delta\}|}{|S_\Delta|} q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right) \\ &= \left(\sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}\right) q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right),\end{aligned}$$

as claimed. □

Proof of Theorem 1.5 (3). By definition,

$$D_q(Y, \ell) = \frac{|\{L \in \mathcal{G}(\mathbb{F}_q) \mid (L \cap Y)(\mathbb{F}_q) = \emptyset\}|}{|\mathcal{G}(\mathbb{F}_q)|}.$$

The condition $(L \cap Y)(\mathbb{F}_q) = \emptyset$ holds if and only if the Frobenius endomorphism fixes no point on the fiber $\pi_{\mathrm{Gr}}^{-1}(L)$. Therefore, over the open subset $\mathcal{G}^\circ \subseteq \mathcal{G}$, Lemma 3.4 yields

$$|\{L \in \mathcal{G}^\circ(\mathbb{F}_q) \mid (L \cap Y)(\mathbb{F}_q) = \emptyset\}| = \left(\sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}\right) q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1/2}\right)\right).$$

The complement $\mathcal{D} := \mathcal{G} \setminus \mathcal{G}^\circ$ is a proper closed subset, so $\dim \mathcal{D} < \dim \mathcal{G}$. Thus, by the Lang–Weil bound [LW54, Theorem 1],

$$|\mathcal{D}(\mathbb{F}_q)| \leq O\left(q^{\dim \mathcal{G} - 1}\right).$$

Comparing the two estimates, the contribution from $\mathcal{D}(\mathbb{F}_q)$ is negligible. On the other hand, the denominator satisfies $|\mathcal{G}(\mathbb{F}_q)| = q^{\dim \mathcal{G}} \left(1 + O\left(q^{-1}\right)\right)$. Therefore,

$$D_q(Y, \ell) = \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!} + O\left(q^{-1/2}\right).$$

In particular,

$$\lim_{q \to \infty} D_q(Y, \ell) = \sum_{i=0}^{\Delta} \frac{(-1)^i}{i!}.$$

This completes the proof. □

4. Quasireflexivity of determinantal varieties

In this section, we prove that determinantal varieties are quasireflexive over fields of arbitrary characteristic. Throughout, we fix the following notation. Let K be an algebraically closed field and fix integers

$$2 \leq \delta \leq n \leq m.$$

We consider the determinantal variety

$$Y_\delta \subseteq \mathbb{P}^N \qquad \text{where} \qquad N = mn - 1$$

consisting of $m \times n$ matrices of rank less than δ. To simplify the notation, we denote its dimension and codimension by

$$\begin{aligned} d &:= \dim Y_\delta = N - (m-\delta+1)(n-\delta+1), \\ c &:= \operatorname{codim} Y_\delta = (m-\delta+1)(n-\delta+1). \end{aligned}$$

By Definition 1.4, to show that Y_δ is quasireflexive, we need to find a c-dimensional linear subspace $L \subseteq \mathbb{P}^N$ that intersects Y_δ in $\deg(Y_\delta) - 1$ points all lying in the smooth locus

$$Y_\delta^{\mathrm{sm}} \subseteq Y_\delta.$$

We begin by treating the hypersurface case.

Proposition 4.1. *If Y_δ is a hypersurface, that is, if $m = n = \delta$, then it is quasireflexive.*

Proof. In this setting, $Y_\delta \subseteq \mathbb{P}^N = \mathbb{P}^{n^2-1}$ is a hypersurface of degree n. It suffices to construct a line $L \subseteq \mathbb{P}^N$ meeting the smooth locus Y_δ^{sm} in $n-1$ distinct points.

Choose $n-1$ distinct points $[a_i : b_i] \in \mathbb{P}^1$, where $i = 1, \ldots, n-1$. When $n = 2$, choose in addition a point $[a_2 : b_2] \neq [a_1 : b_1]$. Consider the morphism

$$\varphi\colon \mathbb{P}^1 \longrightarrow \mathbb{P}^N, \quad \varphi(s:t) = \begin{pmatrix} a_1s+b_1t & a_2s+b_2t & 0 & \cdots & 0 \\ 0 & a_1s+b_1t & 0 & \cdots & 0 \\ 0 & 0 & a_2s+b_2t & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{n-1}s+b_{n-1}t \end{pmatrix}.$$

When $n = 2$, this matrix is interpreted as

$$\begin{pmatrix} a_1s+b_1t & a_2s+b_2t \\ 0 & a_1s+b_1t \end{pmatrix}.$$

Let $L := \varphi(\mathbb{P}^1)$, which is a line in $\mathbb{P}^N$. The scheme-theoretic intersection $L \cap Y_\delta$ is defined by the determinant of $\varphi(s:t)$, which is the polynomial equation

$$(a_1s+b_1t)^2(a_2s+b_2t)\cdots(a_{n-1}s+b_{n-1}t) = 0.$$

This equation has a double root at $[b_1 : -a_1]$ and $n-2$ simple roots at $[b_i : -a_i]$ for $i \geq 2$.

For the matrix $\varphi(b_1 : -a_1)$, the first two diagonal entries vanish, while the $(1,2)$-entry equals $a_2b_1 - a_1b_2 \neq 0$, and all remaining diagonal entries are nonzero. As the matrix has rank $n-1$, it represents a point in the smooth locus Y_δ^{sm}. Since $L \cap Y_\delta$ has degree n, and one intersection point is a smooth point of multiplicity two, the remaining intersection points must also lie in Y_δ^{sm}. Therefore, L meets Y_δ^{sm} in $n-1$ distinct points. This proves that Y_δ is quasireflexive. $\square$

4.1. **Preliminary lemmas on tangent spaces.** The remainder of this section is devoted to the non-hypersurface case. Here, we collect properties of the tangent spaces of a determinantal variety that are needed for our proof.

For any point $P \in Y_\delta^{\mathrm{sm}}$, we write its embedded tangent space as $T_PY_\delta \subseteq \mathbb{P}^N$. Recall that, as shown for example in [Har92, Example 14.16],

$$T_PY_\delta = \left\{ Q \in \mathbb{P}^N \;\middle|\; Q(\ker P) \subseteq \operatorname{im} P \right\}.$$

We will make use of the distinguished smooth point

$$P_0 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & 0 \end{pmatrix} \in Y_\delta^{\mathrm{sm}},$$

where $I_{\delta-1}$ denotes the $(\delta-1)\times(\delta-1)$ identity matrix. Its tangent space $T_{P_0}Y_\delta$ consists of points represented by matrices of the form

$$\begin{pmatrix} *_{\delta-1,\,\delta-1} & *_{\delta-1,\,n-\delta+1} \\ *_{m-\delta+1,\,\delta-1} & 0 \end{pmatrix} \tag{4.1}$$

where $*_{\alpha,\,\beta}$ denotes an arbitrary $\alpha\times\beta$ matrix.

We begin by analyzing the structure of the linear section $Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta$. This section admits a stratification according to the dimension of the intersection $\ker P_0 \cap \ker P \subseteq K^n$. More precisely, we have

$$Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta = \bigcup_{i=0}^{n-\delta+1} T_i$$

where

$$T_i = \{P \in Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta \mid \dim(\ker P_0 \cap \ker P) = (n-\delta+1) - i\}.$$

In particular, since $\ker P_0$ has dimension $n-\delta+1$, we have

$$T_0 = \{P \in Y_\delta^{\mathrm{sm}} \cap T_{P_0}Y_\delta \mid \ker P_0 = \ker P\}.$$

Lemma 4.2. *For each $i \geq 0$, the Zariski closure $\overline{T_i} \subseteq Y_\delta^{\mathrm{sm}}$ is a cone with vertex P_0 and*

$$\begin{aligned} \dim T_i &= i(n-i) + (\delta-1-i)(m+i) - 1 \\ &= d - \left((\delta-1)(n-\delta+1-i) + i(m-n) + 2i^2\right). \end{aligned}$$

In particular, each T_i has positive codimension in Y_δ^{sm}. Moreover, only T_0 can have codimension one, and this occurs if and only if $\delta = n = 2$.

Proof. For a given $P \in T_i$, every point Q on the line spanned by P_0 and P is represented by a linear combination of P_0 and P. For a general such Q, its kernel satisfies the same dimension condition

$$\dim(\ker P_0 \cap \ker Q) = (n-\delta+1) - i.$$

Therefore, the line spanned by P_0 and P is contained in $\overline{T_i}$, showing that $\overline{T_i}$ is a cone with vertex P_0.

To compute the dimension of T_i, we first decompose

$$K^n \cong \ker P_0 \oplus K^{\delta-1}.$$

For each $P \in T_i$, the restrictions of a linear map $\widetilde{P}\colon K^n \to K^m$ representing it to the two summands behave as follows:

- Since $P \in T_{P_0}Y_\delta$, the restriction of $\widetilde{P}$ to $\ker P_0$ defines a linear map

  $$\ker P_0 \cong K^{n-\delta+1} \longrightarrow \operatorname{im} P_0 \cong K^{\delta-1}$$

  of rank i.
- The restriction of $\widetilde{P}$ to the second summand defines a linear map

  $$K^{\delta-1} \longrightarrow K^m$$

  of rank $\delta-1-i$.

Conversely, any linear map $K^n \to K^m$ whose restrictions to the two summands satisfy the above conditions determines a point of T_i. Hence, the dimension of the affine variety underlying T_i is given by the sum of the dimensions of the corresponding affine determinantal varieties, which yields

$$\dim T_i = i(n-i) + (\delta - 1 - i)(m+i) - 1.$$

To determine the codimension of T_i in Y_δ^{sm}, we compute

$$\begin{aligned}\dim Y_\delta - \dim T_i &= (\delta-1)(m+n-\delta+1) - (i(n-i) + (\delta-1-i)(m+i)) \\ &= (\delta-1)(n-\delta+1-i) + i(m-n) + 2i^2.\end{aligned}$$

This quantity is strictly positive since $m \geq n$ and

$$0 \leq i \leq n - \delta + 1,$$

while the equalities $i = 0$ and $i = n - \delta + 1$ cannot hold at the same time. Moreover, it is equal to 1 if and only if $i = 0$ and $\delta = n = 2$. This completes the proof. □

Lemma 4.3. *For a general $P \in Y_\delta^{\mathrm{sm}}$, the tangent space $T_P Y_\delta$ does not contain P_0.*

Proof. Since Y_δ is irreducible and the locus of $P \in Y_\delta^{\mathrm{sm}}$ for which $T_P Y_\delta$ contains P_0 is Zariski closed, it suffices to find a point P such that $T_P Y_\delta$ does not contain P_0. Let $P \in Y_\delta^{\mathrm{sm}}$ be the point represented by the matrix whose (i,i)-entries are equal to 1 for $i = 2, \ldots, \delta$, and zero elsewhere. Let $\{e_1, \ldots, e_n\}$ denote the standard basis for K^n. Then

$$e_1 \in \ker P \quad \text{and} \quad \operatorname{im} P = \operatorname{span}(e_2, \ldots, e_\delta).$$

Since $P_0(e_1) = e_1$, we have

$$P_0(\ker P) \not\subseteq \operatorname{im} P,$$

and hence $P_0 \notin T_P Y_\delta$. This gives an example of P with the desired property. □

Lemma 4.4. *If Y_δ is not a hypersurface, then for a general point $P \in Y_\delta^{\mathrm{sm}}$, we have*

$$\dim\left(T_{P_0} Y_\delta \cap T_P Y_\delta\right) \leq d - 2.$$

Proof. Since Y_δ is irreducible and the function $P \longmapsto \dim\left(T_{P_0} Y_\delta \cap T_P Y_\delta\right)$ is upper semicontinuous on Y_δ, it suffices to find a point P with

$$\dim\left(T_{P_0} Y_\delta \cap T_P Y_\delta\right) \leq d - 2,$$

or equivalently, such that $T_{P_0} Y_\delta \cap T_P Y_\delta$ has codimension at least 2 in $T_{P_0} Y_\delta$. Consider

$$P = \begin{pmatrix} 0 & 0 \\ 0 & I_{\delta-1} \end{pmatrix} \in Y_\delta^{\mathrm{sm}}.$$

Its tangent space $T_P Y_\delta$ consists of points represented by matrices of the form

$$\begin{pmatrix} 0 & *_{m-\delta+1,\,\delta-1} \\ *_{\delta-1,\,n-\delta+1} & *_{\delta-1,\,\delta-1} \end{pmatrix}.$$

Comparing this with (4.1), we see that the intersection $T_{P_0} Y_\delta \cap T_P Y_\delta$ consists of points represented by $m \times n$ matrices with $(m-\delta+1) \times (n-\delta+1)$ zero matrices as the top-left and bottom-right blocks.

- Suppose that
$$m-\delta+1 \leq \delta-1 \qquad \text{or} \qquad n-\delta+1 \leq \delta-1.$$
In this case, the two zero blocks do not overlap. Thus the linear subspace
$$T_{P_0}Y_\delta \cap T_P Y_\delta \subseteq T_{P_0}Y_\delta$$
is defined by $(m-\delta+1)(n-\delta+1)$ independent linear conditions. Hence
$$\operatorname{codim}_{T_{P_0}Y_\delta}\left(T_{P_0}Y_\delta \cap T_P Y_\delta\right) = (m-\delta+1)(n-\delta+1).$$
Note that this quantity equals 1 if and only if $m = n = \delta$.
- Suppose that
$$m-\delta+1 > \delta-1 \qquad \text{and} \qquad n-\delta+1 > \delta-1.$$
In this case, the two zero blocks overlap in a matrix of size
$$(m-2\delta+2) \times (n-2\delta+2).$$
Consequently, the linear subspace $T_{P_0}Y_\delta \cap T_P Y_\delta \subseteq T_{P_0}Y_\delta$ has codimension
$$\begin{aligned}&(m-\delta+1)(n-\delta+1) - (m-2\delta+2)(n-2\delta+2)\\ &= (\delta-1)(m+n-3\delta+3).\end{aligned}$$
This is never equal to 1. Indeed, if it were, then $\delta = 2$ and $m+n = 4$. However, the inequalities above imply $m+n > 4(\delta-1) = 4$, a contradiction.

In both cases, the intersection $T_{P_0}Y_\delta \cap T_P Y_\delta$ has codimension at least 2 in $T_{P_0}Y_\delta$ under the assumption that Y_δ is not a hypersurface. This completes the proof. □

4.2. **Finite linear sections with a tangency.** A linear subspace $L \subseteq \mathbb{P}^N$ is tangent at P_0 if and only if the intersection $L \cap T_{P_0}Y_\delta$ contains a line through P_0. The c-dimensional linear subspaces with this property form the variety
$$X_0 := \left\{ L \in \operatorname{Gr}(\mathbb{P}^c, \mathbb{P}^N) \;\middle|\; L \ni P_0,\ \dim(L \cap T_{P_0}Y_\delta) \geq 1 \right\}.$$
Our goal is to find an $L \in X_0$ that is tangent to Y_δ at P_0 with multiplicity two and transverse at all other intersection points. The proof proceeds in three steps:

(1) There exists $L \in X_0$ that is tangent to Y_δ at P_0 with multiplicity two.
(2) The locus of $L \in X_0$ that meet the singular locus $Y_{\delta-1} \subseteq Y_\delta$ has codimension ≥ 2.
(3) The locus of $L \in X_0$ that are tangent to Y_δ^{sm} at a second point has codimension ≥ 1.

Lemma 4.5. *The variety X_0 is irreducible with $\dim X_0 = cd - 1$.*

Proof. The lines in $T_{P_0}Y_\delta$ passing through P_0 form the variety
$$\mathcal{B} := \left\{ \ell \in \operatorname{Gr}(\mathbb{P}^1, T_{P_0}Y_\delta) \;\middle|\; P_0 \in \ell \right\} \cong \mathbb{P}^{d-1}.$$
Consider the incidence correspondence
$$\begin{tikzcd} & I_0 := \{(L,\ell) \in X_0 \times \mathcal{B} \mid L \supseteq \ell\} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ X_0 & & \mathcal{B}. \end{tikzcd}$$

For each $\ell \in \mathcal{B}$, the fiber $\pi_2^{-1}(\ell) \subseteq I_0$ consists of all pairs (L, ℓ) such that $L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N)$ contains ℓ. This gives I_0 the structure of a bundle over $\mathcal{B}$ with fiber $\mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2})$. It follows that I_0 is irreducible with

$$\dim I_0 = \dim \mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2}) + \dim \mathcal{B} = (c-1)d + (d-1) = cd - 1.$$

Moreover, a general $L \in X_0$ intersects $T_{P_0} Y_\delta$ in exactly one line. Hence π_1 is birational. It follows that X_0 is irreducible with $\dim X_0 = \dim I_0 = cd - 1$. $\square$

Lemma 4.6. *There exists $L \in X_0$ that is tangent to Y_δ at P_0 with multiplicity two.*

Proof. We work in the affine chart of $\mathbb{P}^N$ consisting of $m \times n$ matrices whose $(1,1)$-entry is equal to 1. Let $L \subseteq X_0$ be the linear subspace whose elements in this chart have the form

$$M := \begin{pmatrix} I_{\delta-1} & A \\ B & C \end{pmatrix}$$

where

- $A = (a_{ij})$ has $a_{ij} = 0$ for all $(i,j) \neq (1,1)$,
- $B = (b_{ij})$ has $b_{11} = a_{11}$ and $b_{ij} = 0$ for all $(i,j) \neq (1,1)$,
- $C = (c_{ij})$ has $c_{11} = 0$.

By the standard block row-reduction,

$$\mathrm{rank}(M) = \mathrm{rank} \begin{pmatrix} I_{\delta-1} & A \\ 0 & C - BA \end{pmatrix} = (\delta - 1) + \mathrm{rank}(C - BA).$$

Hence, $M \in L \cap Y_\delta$ if and only if $C = BA$, which holds if and only if

$$b_{11} a_{11} = a_{11}^2 = 0 \qquad \text{and} \qquad c_{ij} = 0 \quad \text{for all} \quad (i,j).$$

The equations $c_{ij} = 0$ define a line in L passing through P_0. On this line, $a_{11}^2 = 0$ defines a point of multiplicity two at P_0. Hence, L is a linear subspace with the desired properties. $\square$

Lemma 4.7. *The locus $X_s \subseteq X_0$ of linear subspaces $L \in X_0$ that meet the singular locus $Y_{\delta-1} \subseteq Y_\delta$ has dimension at most $cd - 3$.*

Proof. Consider the incidence correspondence

$$\begin{tikzcd} & I_s := \{(L, P) \in X_s \times Y_{\delta-1} \mid L \ni P\} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ X_s & & Y_{\delta-1}. \end{tikzcd}$$

For each singular point $P \in Y_{\delta-1}$, every $L \in X_s$ containing P must also contain the line spanned by P_0 and P. Hence, if nonempty, the fiber $\pi_2^{-1}(P)$ can be identified with a subset of $\mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2})$. Therefore,

$$\begin{aligned} \dim I_s &\leq \dim Y_{\delta-1} + \dim \mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2}) \\ &= (N - (m - \delta + 2)(n - \delta + 2)) + (c-1)(N-c) \\ &= (d - (m + n - 2\delta + 3)) + (c-1)d \\ &= cd - (m + n - 2\delta + 3) \\ &\leq cd - 3. \end{aligned}$$

By the definition of X_s, the projection π_1 is surjective. Thus $\dim X_s \leq \dim I_s \leq cd - 3$. $\square$

4.3. **Finite linear sections with extra tangency.** Our next goal is to bound the dimension of the locus $X_t \subseteq X_0$ defined by

$$X_t := \left\{ L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N) \;\middle|\; L \text{ is tangent at } P_0 \text{ and some } P \in Y_\delta^{\mathrm{sm}} \setminus P_0 \right\}.$$

Lemma 4.8. *The locus $X_t' \subseteq X_t$ defined by*

$$X_t' := \{ L \in X_t \mid P \in T_{P_0} Y_\delta \}$$

has dimension at most $cd - 2$.

Proof. For each $P \in Y_\delta^{\mathrm{sm}} \setminus P_0$, the subspaces $L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N)$ containing both P_0 and P are precisely those that contain the line spanned by P_0 and P. Such subspaces are parametrized by $\mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2})$. By Lemma 4.2, the intersection $Y_\delta^{\mathrm{sm}} \cap T_{P_0} Y_\delta$ decomposes into a union of components, each of which is a cone with vertex P_0 and has dimension at most $d-1$. The cone structure implies that for any $P \in Y_\delta^{\mathrm{sm}} \cap T_{P_0} Y_\delta$ with $P \neq P_0$, all points on the line spanned by P_0 and P determine the same family of subspaces $L \in \mathrm{Gr}(\mathbb{P}^c, \mathbb{P}^N)$ containing both points. Consequently,

$$\begin{aligned} \dim X_t' &\leq \dim \mathrm{Gr}(\mathbb{P}^{c-2}, \mathbb{P}^{N-2}) + (d-1) - 1 \\ &= (c-1)d + d - 2 \\ &= cd - 2. \end{aligned}$$

This yields the desired bound. □

It remains to analyze the open complement

$$X_t^\circ := X_t \setminus X_t' = \{ L \in X_t \mid P \notin T_{P_0} Y_\delta \}.$$

For each $L \in X_t$, there exist lines $\ell_0 \subseteq T_{P_0} Y_\delta$ and $\ell \subseteq T_P Y_\delta$ such that

$$P_0 \in \ell_0 \subseteq L \cap T_{P_0} Y_\delta \qquad \text{and} \qquad P \in \ell \subseteq L \cap T_P Y_\delta. \tag{4.2}$$

In view of this, we define

$$\begin{aligned} \mathcal{B}_0 &:= \left\{ \ell_0 \in \mathrm{Gr}(\mathbb{P}^1, T_{P_0} Y_\delta) \;\middle|\; P_0 \in \ell_0 \right\} \cong \mathbb{P}^{d-1}, \\ \mathcal{B}_1 &:= \left\{ \ell \in \mathrm{Gr}(\mathbb{P}^1, T_P Y_\delta) \;\middle|\; P \in \ell \right\} \cong \mathbb{P}^{d-1}, \end{aligned}$$

and consider the incidence correspondence

$$I_t \subseteq X_t^\circ \times Y_\delta^{\mathrm{sm}} \times \mathcal{B}_0 \times \mathcal{B}_1,$$

consisting of quadruples (L, P, ℓ_0, ℓ) satisfying (4.2). Each such quadruple satisfies:

- The lines ℓ_0 and ℓ are distinct.
- If ℓ_0 and ℓ intersect, the intersection point is different from P.

Indeed, if either of these conditions fails, then $P \in \ell_0 \subseteq T_{P_0} Y_\delta$, contradicting the assumption that $L \in X_t^\circ$. Hence I_t decomposes as a disjoint union

$$I_t = I_t' \cup I_t'',$$

where

$$\begin{aligned} I_t' &= \{ (L, P, \ell_0, \ell) \in I_t \mid \ell_0 \cap \ell = \emptyset \}, \\ I_t'' &= \{ (L, P, \ell_0, \ell) \in I_t \mid \ell_0 \cap \ell = \{Q\} \text{ for some } Q \neq P \}. \end{aligned}$$

Lemma 4.9. $\dim I_t' \leq cd - 2$.

Proof. Consider the projection map

$$\pi' \colon I'_t \longrightarrow Y_\delta \times \mathcal{B}_0 \times \mathcal{B}_1 : (L, P, \ell_0, \ell) \longmapsto (P, \ell_0, \ell).$$

For each quadruple $(L, P, \ell_0, \ell) \in I'_t$, the condition $\ell_0 \cap \ell = \emptyset$ implies that their span

$$\Pi := \langle \ell_0, \ell \rangle \subseteq \mathbb{P}^N$$

is a 3-plane. Any L occurring in such a quadruple must contain Π. Hence, over a triple

$$(P, \ell_0, \ell) \in Y_\delta \times \mathcal{B}_0 \times \mathcal{B}_1,$$

the fiber $(\pi')^{-1}(P, \ell_0, \ell)$ is contained in $\mathrm{Gr}(\mathbb{P}^{c-4}, \mathbb{P}^{N-4})$. It follows that

$$\begin{aligned}
\dim I'_t &\leq \dim Y_\delta + \dim \mathcal{B}_0 + \dim \mathcal{B}_1 + \dim \mathrm{Gr}(\mathbb{P}^{c-4}, \mathbb{P}^{N-4}) \\
&= d + (d-1) + (d-1) + (c-3)(N-c) \\
&= 3d - 2 + (c-3)d \\
&= cd - 2.
\end{aligned}$$

This proves the desired upper bound. □

Lemma 4.10. $\dim I''_t \leq cd - 2$, *under the assumption that* Y_δ *is not a hypersurface.*

Proof. Assigning to each $(L, P, \ell_0, \ell) \in I''_t$ the intersection point $\ell_0 \cap \ell$ defines a morphism

$$\iota \colon I''_t \longrightarrow \mathbb{P}^N : (L, P, \ell_0, \ell) \longmapsto \ell_0 \cap \ell.$$

Let J_0 denote the fiber over P_0 and J_1 its complement. That is,

$$J_0 := \iota^{-1}(P_0), \qquad J_1 := I''_t \setminus J_0.$$

We first bound the dimension of J_0. Fix a point $P \in Y_\delta^{\mathrm{sm}}$ in the image of the projection

$$\pi \colon J_0 \longrightarrow Y_\delta^{\mathrm{sm}} : (L, P, \ell_0, \ell) \longmapsto P.$$

By the defining condition of I''_t, we have $P_0 = \ell_0 \cap \ell \neq P$. Thus, for any $(L, P, \ell_0, \ell) \in \pi^{-1}(P)$,

- ℓ is uniquely determined as the line through P_0 and P,
- ℓ_0 is a general line in $T_{P_0} Y_\delta$ passing through P_0, and
- L is a c-dimensional linear subspace containing the plane spanned by ℓ_0 and ℓ.

These imply that

$$\dim \pi^{-1}(P) \leq \dim \mathrm{Gr}(\mathbb{P}^{c-3}, \mathbb{P}^{N-3}) + (d-1) = (c-2)d + (d-1).$$

Moreover, we have $P_0 \in \ell \subseteq T_P Y_\delta$, so in particular $T_P Y_\delta$ contains P_0. By Lemma 4.3, the locus of such points $P \in Y_\delta^{\mathrm{sm}}$ has dimension at most $d-1$. Consequently,

$$\dim J_0 \leq \dim \pi^{-1}(P) + (d-1) \leq (c-2)d + 2(d-1) = cd - 2.$$

Next, we bound the dimension of J_1. Consider the projection

$$\rho \colon J_1 \longrightarrow Y_\delta^{\mathrm{sm}} : (L, P, \ell_0, \ell) \longmapsto P.$$

For each $P \in Y_\delta^{\mathrm{sm}}$, the restriction of ι to the fiber $\rho^{-1}(P) \subseteq J_1$ defines a map

$$\iota_P := \iota|_{\rho^{-1}(P)} \colon \rho^{-1}(P) \longrightarrow T_{P_0} Y_\delta \cap T_P Y_\delta \cong \mathbb{P}^e.$$

By the defining conditions of I''_t and J_1, we have

$$P_0 \neq \ell_0 \cap \ell \neq P.$$

Thus, for each $Q \in \mathrm{im}(\iota_P)$, the fiber $\iota_P^{-1}(Q)$ consists of $(L, P, \ell_0, \ell) \in \rho^{-1}(P)$ such that

- ℓ_0 is uniquely determined as the line through P_0 and Q,
- ℓ is uniquely determined as the line through P and Q, and
- L is a c-dimensional linear subspace containing the plane spanned by ℓ_0 and ℓ.

Moreover, since Y_δ is not a hypersurface, Lemma 4.4 implies $e \leq d-2$. Hence,

$$\dim \rho^{-1}(P) \leq \dim \operatorname{Gr}(\mathbb{P}^{c-3}, \mathbb{P}^{N-3}) + e \leq (c-2)d + (d-2).$$

Consequently,

$$\dim J_1 \leq \dim \rho^{-1}(P) + d \leq (c-2)d + (d-2) + d = cd - 2.$$

It follows that

$$\dim I_t'' \leq \max\{\dim J_0,\, \dim J_1\} \leq cd - 2,$$

which proves the desired bound. □

Lemma 4.11. $\dim X_t \leq cd-2$, *under the assumption that* Y_δ *is not a hypersurface.*

Proof. Due to Lemma 4.8, it remains to prove that

$$\dim X_t^\circ \leq cd - 2.$$

Each $L \in X_t^\circ$ is tangent at both P_0 and some point $P \in Y_\delta^{\mathrm{sm}} \setminus P_0$. Hence there exist lines

$$\ell_0 \subseteq L \cap T_{P_0} Y_\delta \qquad \text{and} \qquad \ell \subseteq L \cap T_P Y_\delta$$

passing through P_0 and P, respectively. This shows that the projection map

$$I_t \longrightarrow X_t^\circ : (L, P, \ell_0, \ell) \longmapsto L$$

is surjective. It follows that

$$\dim X_t^\circ \leq \dim I_t \leq \max\{\dim I_t',\, \dim I_t''\},$$

which is at most $cd-2$ by Lemmas 4.9 and 4.10. □

Theorem 4.12. *Every determinantal variety* $Y_\delta \subseteq \mathbb{P}^N$ *is quasireflexive. That is, there exists a linear subspace* $L \subseteq \mathbb{P}^N$ *of dimension* $N - \dim Y_\delta$ *that intersects* Y_δ *in* $\deg(Y_\delta) - 1$ *points at which* Y_δ *is smooth.*

Proof. The hypersurface case is proved in Proposition 4.1. For non-hypersurfaces, Lemma 4.6 implies the existence of a Zariski open subset $U \subseteq X_0$ parametrizing linear subspaces whose tangency with Y_δ at P_0 has multiplicity two. By Lemmas 4.5, 4.7 and 4.11, the locus in X_0 parametrizing linear subspaces that meet Y_δ non-transversely at points other than P_0 form a proper closed subset. Therefore, U contains a member that is tangent to Y_δ at P_0 with multiplicity two and intersects Y_δ transversely at all other points. □

Appendix A. Reflexivity of determinantal varieties

For curves over a field of characteristic not equal to 2, reflexivity and quasireflexivity are equivalent notions by [Ent21, Proposition 2.1]. In higher dimensions, following the second paragraph of the proof of [BH86, Theorem at p. 906], one deduces that:

Proposition A.1. *Every reflexive variety over a field of characteristic not equal to 2 is quasireflexive.*

In what follows, we prove that determinantal varieties defined over a field of arbitrary characteristic are reflexive. By Proposition A.1, this already implies their quasireflexivity in characteristic different from 2. However, it does not address the case of characteristic 2, which is of particular importance in coding theory.

We begin by recalling the definition of reflexivity. Let $Y \subseteq \mathbb{P}^N$ be a projective variety defined over an algebraically closed field K of arbitrary characteristic. Its *conormal variety*

$$C(Y) \subseteq \mathbb{P}^N \times \left(\mathbb{P}^N\right)^*$$

is the Zariski closure of the set of pairs $(P, H) \in \mathbb{P}^N \times \left(\mathbb{P}^N\right)^*$ where

- P belongs to the smooth locus $Y^{\mathrm{sm}} \subseteq Y$, and
- H is a hyperplane in $\mathbb{P}^N$ containing the tangent space $T_P Y \subseteq \mathbb{P}^N$.

Consider the two projection maps

$$\begin{tikzcd} & C(Y) \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ Y & & \left(\mathbb{P}^N\right)^* . \end{tikzcd}$$

The *dual variety* of Y is defined as the image of the second projection

$$Y^* := \pi_2(C(Y)) \subseteq \left(\mathbb{P}^N\right)^*,$$

and Y is *reflexive* if $C(Y) \cong C(Y^*)$ under the natural isomorphisms

$$\mathbb{P}^N \times \left(\mathbb{P}^N\right)^* \cong \left(\mathbb{P}^N\right)^* \times \mathbb{P}^N \cong \left(\mathbb{P}^N\right)^* \times \left(\mathbb{P}^N\right)^{**}. \tag{A.1}$$

Let $Y_\delta \subseteq \mathbb{P}^N = \mathbb{P}^{mn-1}$ be the determinantal variety over K consisting of $m \times n$ matrices, where $m \geq n$, of rank $< \delta$. Recall that the tangent space at a smooth point $P \in Y_\delta^{\mathrm{sm}}$ is given by

$$T_P Y_\delta = \left\{ Q \in \mathbb{P}^N \;\middle|\; Q(\ker P) \subseteq \operatorname{im} P \right\}.$$

On the other hand, there is a nondegenerate bilinear form

$$\langle -, - \rangle \colon K^{mn} \times K^{mn} \longrightarrow K, \quad (A, B) \longmapsto \operatorname{tr}(AB^T).$$

In coordinates, if we write $A = (a_{ij})$ and $B = (b_{ij})$, then

$$\langle A, B \rangle = \sum_{i,j} a_{ij} b_{ij}.$$

We identify $(\mathbb{P}^N)^*$ with $\mathbb{P}^N$ via the induced isomorphism, which is explicitly given by

$$\mathbb{P}^N \xrightarrow{\sim} (\mathbb{P}^N)^* : P \longmapsto H_P = \left\{ \textstyle\sum_{i,j} p_{ij} x_{ij} = 0 \right\}$$

where (p_{ij}) is any matrix representing P.

Lemma A.2. *Let $I_{\delta-1}$ denote the $(\delta - 1) \times (\delta - 1)$ identity matrix. The fiber over*

$$P_0 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & 0 \end{pmatrix} \in Y_\delta$$

under the projection $\pi_1 \colon C(Y_\delta) \longrightarrow Y_\delta$ has the form

$$\pi_1^{-1}(P_0) = \left\{ \left(P_0, \begin{pmatrix} 0 & 0 \\ 0 & *_{m-\delta+1,\, n-\delta+1} \end{pmatrix} \right) \right\}$$

where $*_{\alpha,\beta}$ denotes an arbitrary $\alpha \times \beta$ matrix. In particular,

$$\pi_2\left(\pi_1^{-1}(P_0)\right) \subseteq Y_{n-\delta+2}.$$

Proof. Every $Q \in T_{P_0}Y_\delta$ is represented by a matrix of the form

$$\begin{pmatrix} *_{\delta-1,\,\delta-1} & *_{\delta-1,\,n-\delta+1} \\ *_{m-\delta+1,\,\delta-1} & 0 \end{pmatrix}.$$

Hence, H_P contains $T_{P_0}Y_\delta$ if and only if P is represented by

$$\widetilde{P} = \begin{pmatrix} 0 & 0 \\ 0 & *_{m-\delta+1,\,n-\delta+1} \end{pmatrix}.$$

Since $(P_0, P) \in \mathbb{P}^N \times (\mathbb{P}^N)^*$ belongs to $\pi_1^{-1}(P_0)$ if and only if $T_{P_0}Y_\delta \subseteq H_P$, this proves the first statement. The second statement follows immediately from the expression of $\widetilde{P}$. $\square$

Lemma A.3. *For any $A \in \mathrm{GL}_m(K)$ and $B \in \mathrm{GL}_n(K)$,*

$$T_PY_\delta \subseteq H_Q \iff T_{APB}Y_\delta \subseteq H_{A^\dagger QB^\dagger}$$

where $A^\dagger = (A^{-1})^T$ and $B^\dagger = (B^{-1})^T$.

Proof. Note that

$$\mathrm{tr}\left(P\,Q^T\right) = \mathrm{tr}\left((APB)\,(A^\dagger QB^\dagger)^T\right). \tag{A.2}$$

Using the identities

- $\ker(APB) = \ker(PB) = B^{-1}\ker P$,
- $\mathrm{im}(APB) = \mathrm{im}(AP) = A\ \mathrm{im}\,P$,

we deduce for any $M \in \mathrm{Hom}(K^n, K^m)$ that

$$\begin{aligned}(AMB)\ \ker(APB) \subseteq \mathrm{im}(APB) &\iff (AMB)\,(B^{-1}\ker P) \subseteq A\ \mathrm{im}\,P \\ &\iff M\ \ker P \subseteq \mathrm{im}\,P.\end{aligned}$$

It follows that

$$A\,(T_PY_\delta)\,B = T_{APB}Y_\delta. \tag{A.3}$$

Consequently,

$$\begin{aligned} T_PY_\delta \subseteq H_Q &\iff \langle T_PY_\delta,\, Q\rangle = 0 \\ \text{(by (A.2))} \quad &\iff \left\langle A\,(T_PY_\delta)\,B,\, A^\dagger QB^\dagger\right\rangle = 0 \\ \text{(by (A.3))} \quad &\iff \left\langle T_{APB}Y_\delta,\, A^\dagger QB^\dagger\right\rangle = 0 \iff T_{APB}Y_\delta \subseteq H_{A^\dagger QB^\dagger}. \end{aligned}$$

This completes the proof. $\square$

Lemma A.4. *For any $(P, Q) \in Y_\delta^{\mathrm{sm}} \times Y_{n-\delta+2}^{\mathrm{sm}}$, the inclusion $T_PY_\delta \subseteq H_Q$ holds if and only if there exist matrices $A \in \mathrm{GL}_m(K)$ and $B \in \mathrm{GL}_n(K)$ such that*

$$APB = P_0, \qquad A^\dagger QB^\dagger = \begin{pmatrix} 0 & 0 \\ 0 & I_{n-\delta+1} \end{pmatrix},$$

where $A^\dagger = (A^{-1})^T$ and $B^\dagger = (B^{-1})^T$.

Proof. If there exist matrices A and B with the given identities, then

$$T_{APB}Y_\delta = T_{P_0}Y_\delta \subseteq H_{A^\dagger Q B^\dagger},$$

which implies $T_P Y_\delta \subseteq H_Q$ by Lemma A.3.

It remains to prove the converse. Since P has rank $\delta - 1$, there exist $A_1 \in \mathrm{GL}_m(K)$ and $B_1 \in \mathrm{GL}_n(K)$ such that $A_1 P B_1 = P_0$. By Lemma A.3,

$$T_P Y_\delta \subseteq H_Q \iff T_{A_1 P B_1} Y_\delta = T_{P_0} Y_\delta \subseteq H_{A_1^\dagger Q B_1^\dagger}.$$

The last inclusion relation $T_{P_0} Y_\delta \subseteq H_{A_1^\dagger Q B_1^\dagger}$, together with Lemma A.2, implies

$$A_1^\dagger Q B_1^\dagger = \begin{pmatrix} 0 & 0 \\ 0 & C \end{pmatrix}$$

for some $(m-\delta+1) \times (n-\delta+1)$ matrix C. Since $\operatorname{rank} C = \operatorname{rank} Q = n - \delta + 1$, there exist $A' \in \mathrm{GL}_{m-\delta+1}(K)$ and $B' \in \mathrm{GL}_{n-\delta+1}(K)$ such that

$$A'^\dagger C B'^\dagger = \begin{pmatrix} 0 \\ I_{n-\delta+1} \end{pmatrix}_{(m-\delta+1)\times(n-\delta+1)}.$$

Now consider the matrices

$$A_2 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & A' \end{pmatrix} \in \mathrm{GL}_m(K) \quad \text{and} \quad B_2 = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & B' \end{pmatrix} \in \mathrm{GL}_n(K).$$

Define $A = A_2 A_1$ and $B = B_1 B_2$. Then they satisfy the desired property:

$$APB = \begin{pmatrix} I_{\delta-1} & 0 \\ 0 & 0 \end{pmatrix}, \qquad A^\dagger Q B^\dagger = \begin{pmatrix} 0 & 0 \\ 0 & I_{n-\delta+1} \end{pmatrix}.$$

This completes the proof. □

Proposition A.5. *For any* $(P, Q) \in Y_\delta^{\mathrm{sm}} \times Y_{n-\delta+2}^{\mathrm{sm}}$,

$$T_P Y_\delta \subseteq H_Q \iff T_Q Y_{n-\delta+2} \subseteq H_P.$$

In particular, Y_δ *has dual variety* $Y_{n-\delta+2}$ *and is reflexive.*

Proof. The statement of Lemma A.4 is symmetric in P and Q up to permutations of rows and columns, since $n - (n-\delta+2) + 2 = \delta$. Therefore, the first statement holds. This implies that Y_δ and $Y_{n-\delta+2}$ are dual to each other, and that there is an isomorphism

$$\left\{(P,Q) \in Y_\delta^{\mathrm{sm}} \times Y_{n-\delta+2}^{\mathrm{sm}} \;\middle|\; T_P Y_\delta \subseteq H_Q\right\} \cong \left\{(Q,P) \in Y_{n-\delta+2}^{\mathrm{sm}} \times Y_\delta^{\mathrm{sm}} \;\middle|\; T_Q Y_{n-\delta+2} \subseteq H_P\right\}.$$

Taking the Zariski closures of both sides yields an isomorphism

$$C(Y_\delta) \cong C(Y_{n-\delta+2}),$$

which is compatible with isomorphism (A.1). This shows that Y_δ is reflexive. □

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